Legged Robots Practical

17.10.2023

Plan

Single Leg

- **W1:** Introduction
- **W2:** Derivation of double pendulum's kinematics and dynamics
- **W3:** Jacobian (Cartesian PD + Force Control)
- **W4:** Inverse Kinematics (compare with force control)
- **W5:** Single-leg hopping

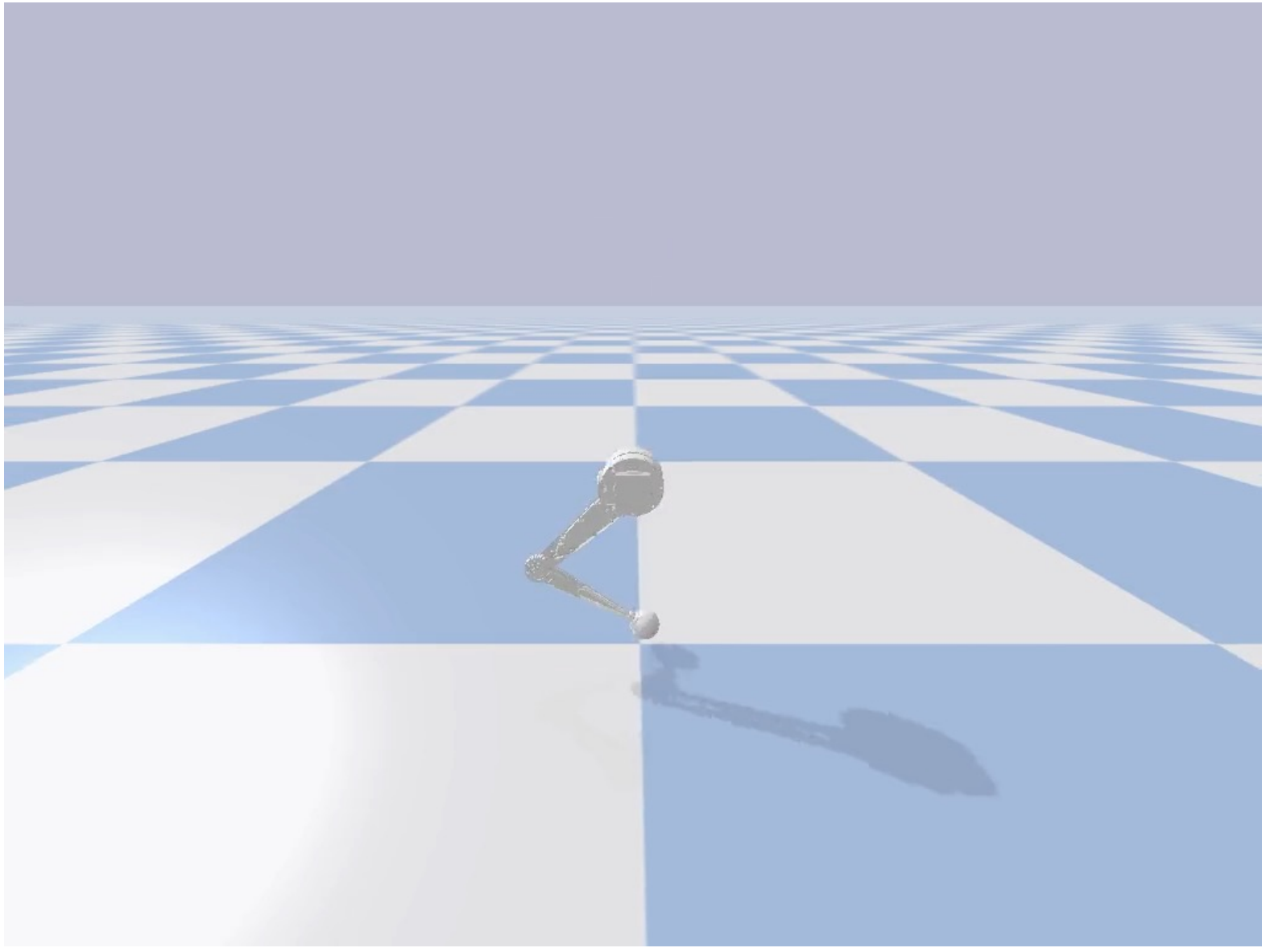
Biped

- **W6-8:** Biped Walking with Divergent Component of Motion (BD Atlas) **[Mini-Report]**

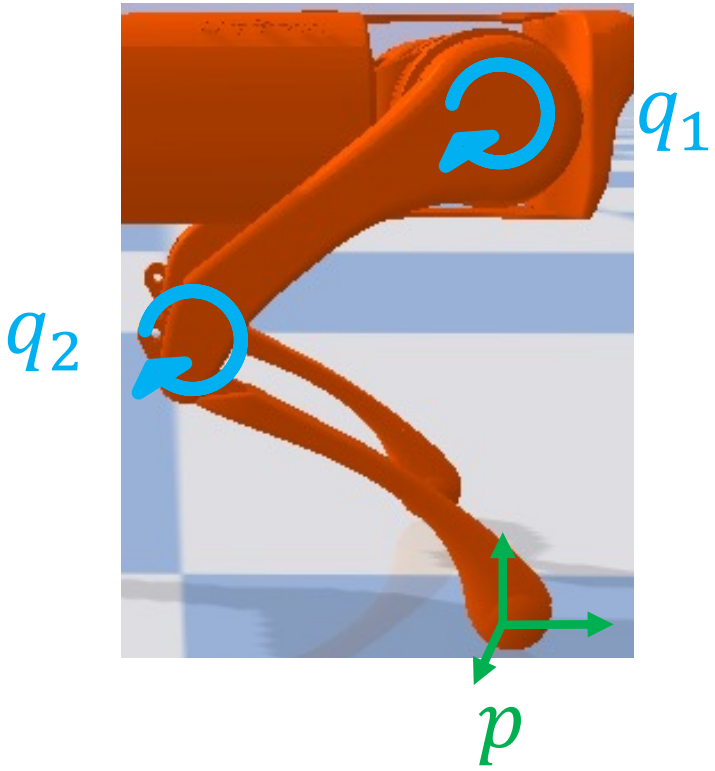
Quadruped

- **W9-10:** Quadruped CPG Trot
- **W11-14:** Quadruped Locomotion Project (CPGs, Deep RL) **[Report - Quadruped]**

Single-Leg Hopping



Joint angles ←→ Cartesian space



$$p = f(q)$$ Forward kinematics

$$q = f^{-1}(p)$$ Inverse kinematics

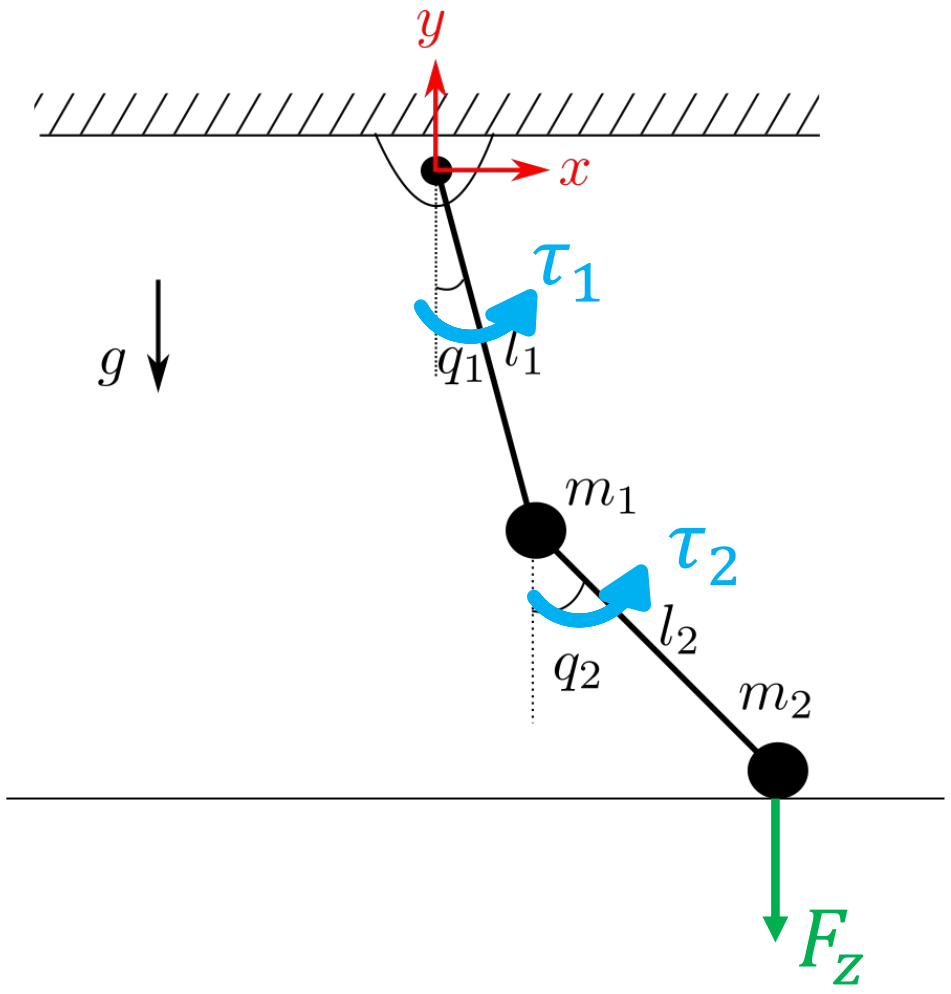
$$\dot{p} = v = J(q)\dot{q}$$ Foot linear velocity

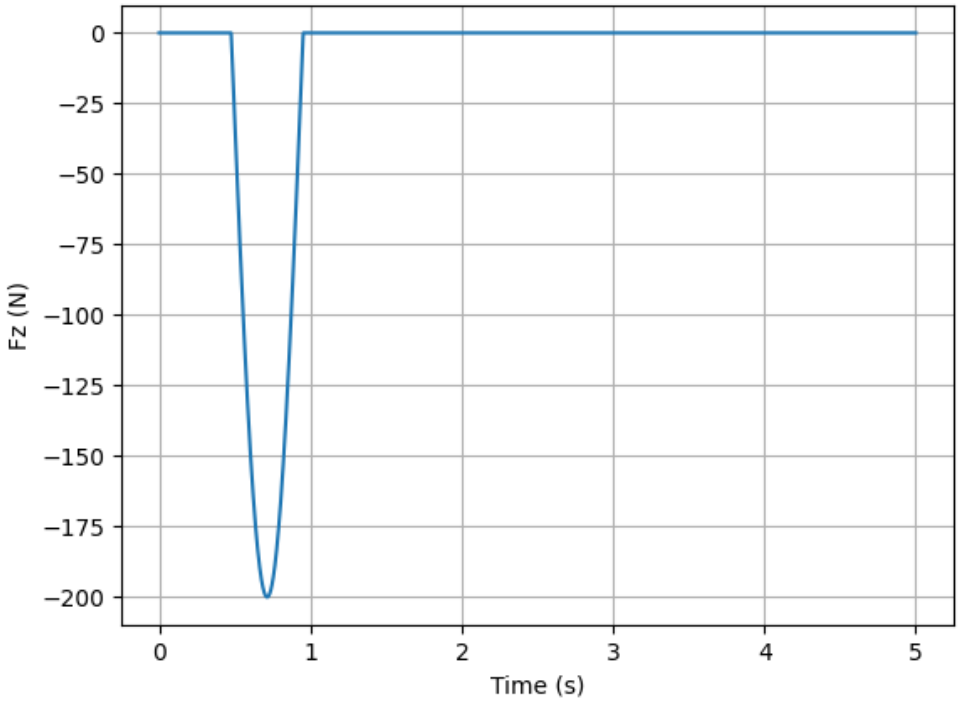
$$\tau = J^T(q)F$$ Map desired end effector force to torques

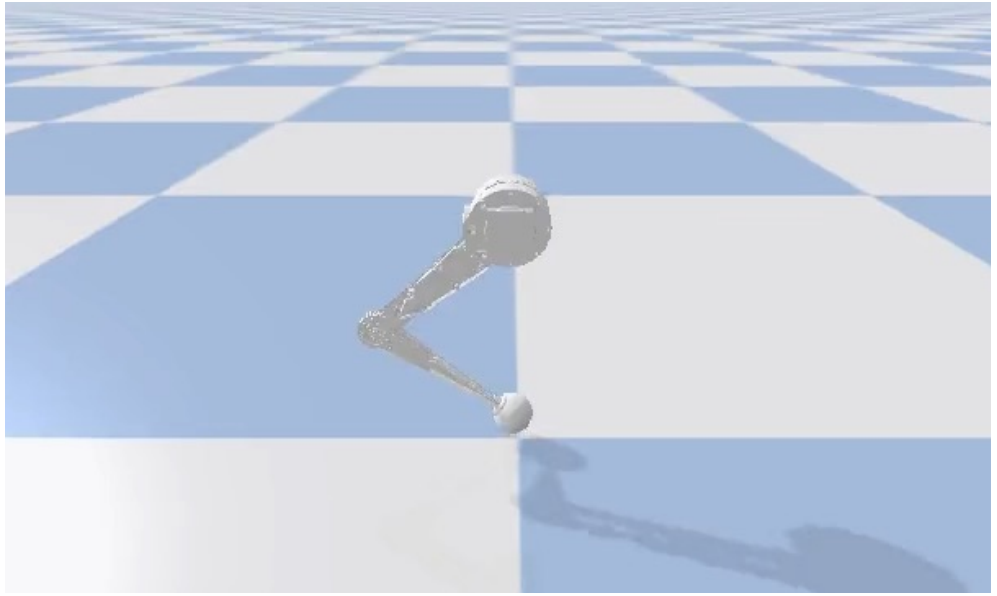
How should we design a jumping motion?

1. Joint space?
2. Task space?
3. Force?

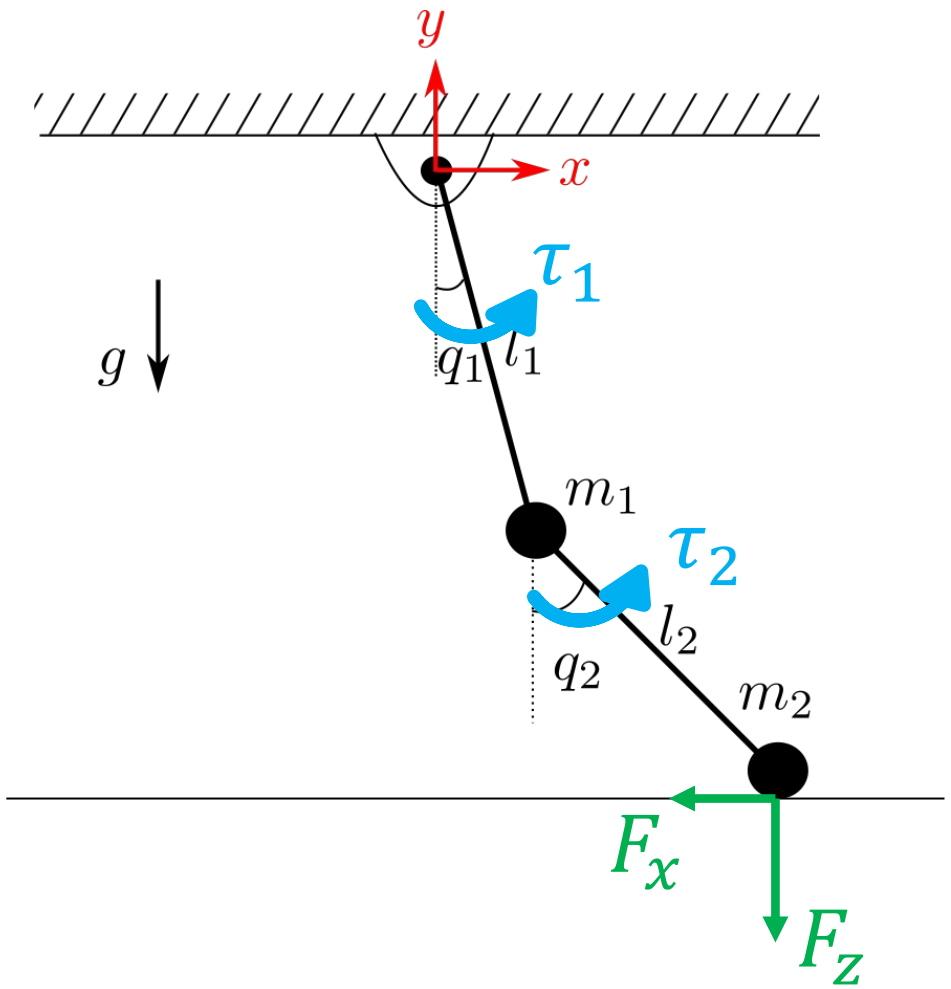
Force Profile: Single Jump

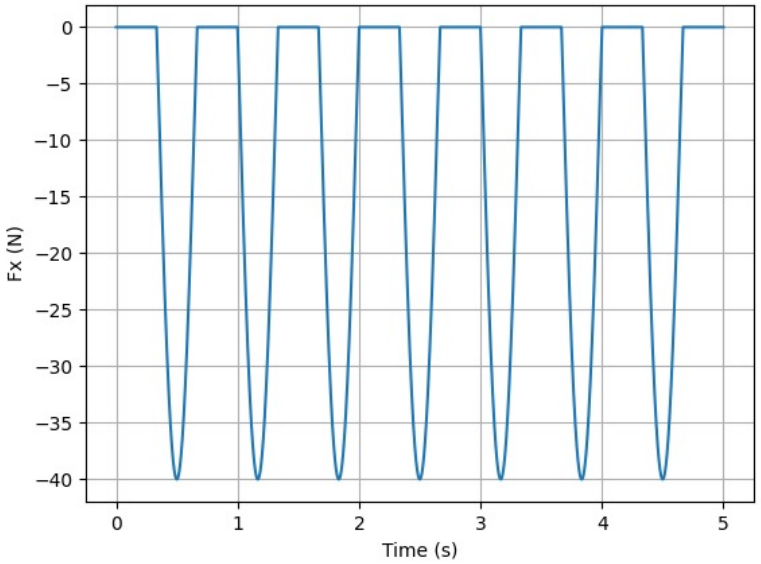


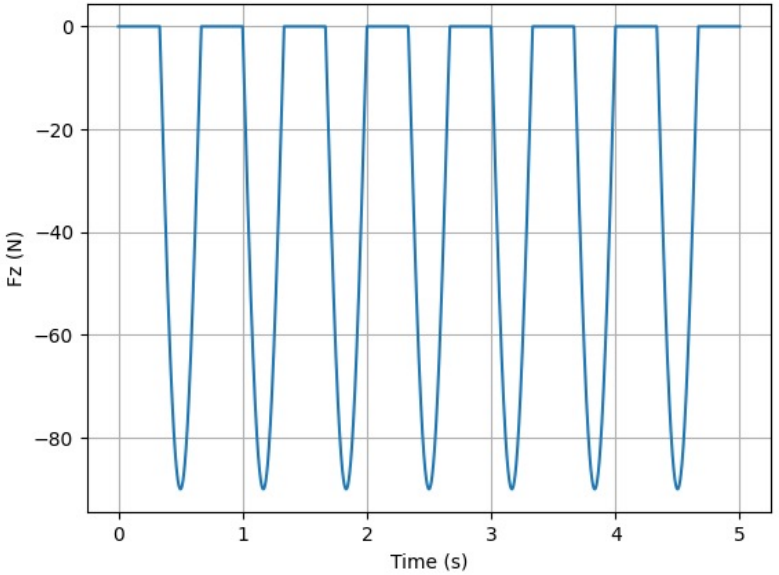


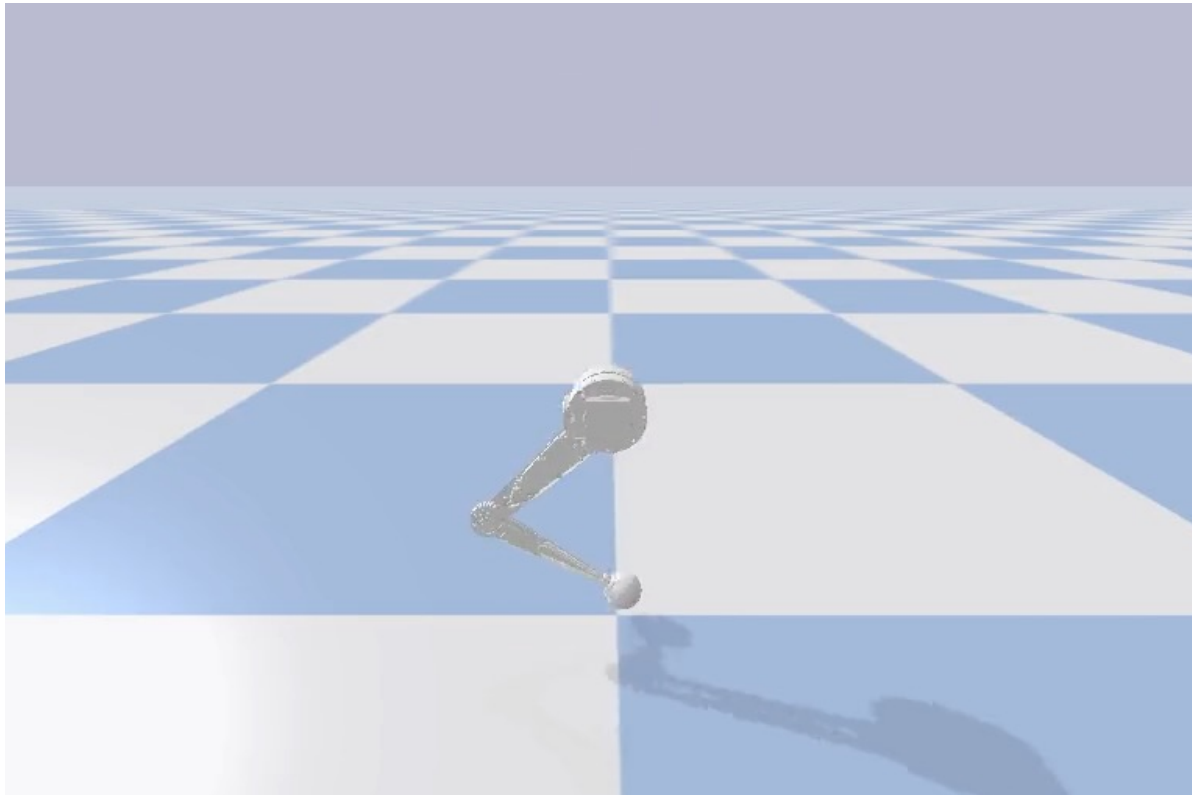


Force Profile: Continuous Jumping



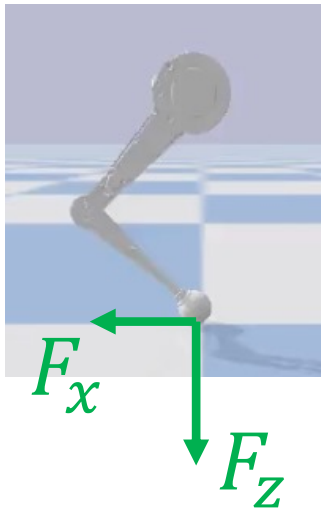


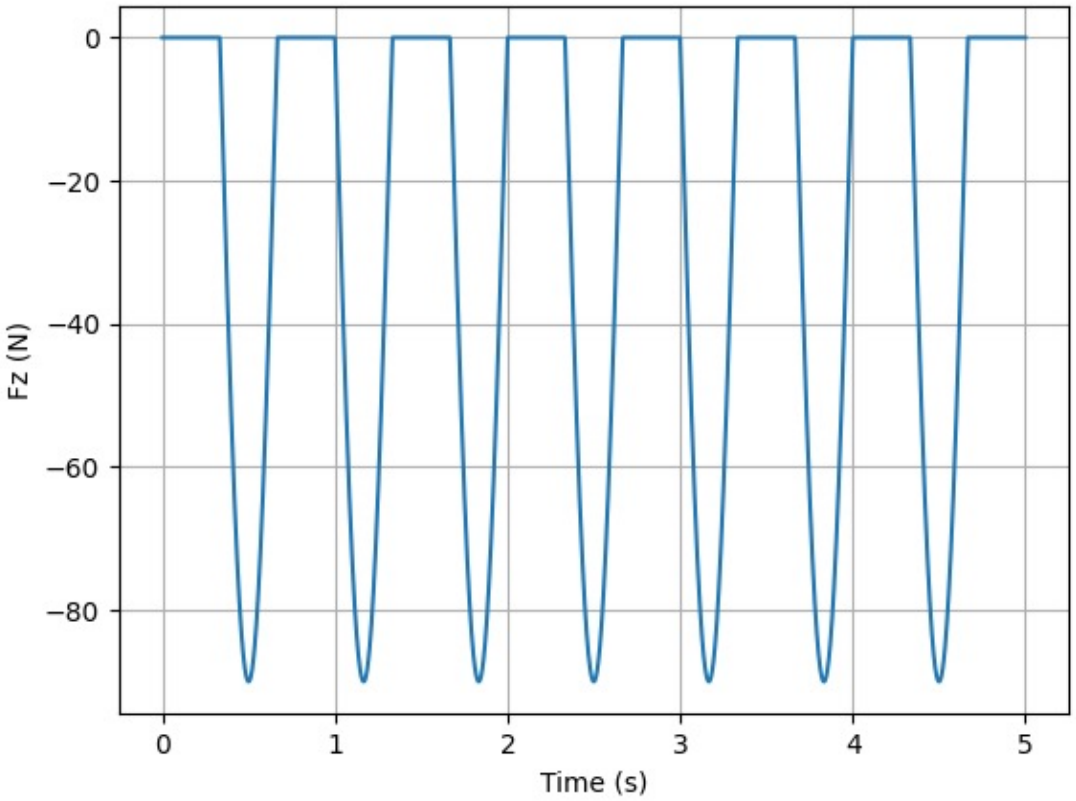




How should we design the profile?

- Design decisions
  - Peak force?
  - Frequency?
  - Initial/nominal position?
  - Joint/Cartesian gains?
    - (In swing/stance)?





$$\tau_{joint} = K_{p,joint}(q_d - q) + K_{d,joint}(\dot{q}_d - \dot{q})$$ Joint PD

$$\tau_{Cartesian} = J^T(q)\left[K_{p,Cartesian}(p_d - p) + K_{d,Cartesian}(v_d - v)\right]$$ Cartesian PD

$$\tau_{final} = \tau_{joint} + \tau_{Cartesian} + J^T(q)F$$ Total applied torque

Optimization

- Choose parameters you would like to optimize

$$\boldsymbol{x} = [f, F_{z_{\text{peak}}}, \dots]$$

- Formulate an unconstrained optimization with objective ($-\,max_height$?)

$$\underset{\boldsymbol{x}}{\text{minimize}}\ \boldsymbol{f}(\boldsymbol{x})$$

- Set upper/lower bounds on variables, and give an initial condition
- We provide an interface to *pymoo* and CMA-ES

https://pymoo.org/index.html
https://pymoo.org/algorithms/soo/cmaes.html

Jump Around

- Please merge your code with https://gitlab.epfl.ch/bellegar/lr-leg
- Upload your working code in Assignment 04 (1 per team)
- Today's assignment is not graded